

Bayesian Spatial Search - Deliverable 2

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1 Prior distribution

The prior $\pi_j = \mathbb{P}(Z = j)$ is built from the physics of the accident and the two witness statements:

$$Z = x_E + \alpha v_{\text{plane}} + \tau (v_{\text{wind}} + v_{\text{drift}}) + 1.5\varepsilon + \eta \hat{n}_{\perp}.$$

Justification. The error term $\varepsilon \sim \mathcal{N}(0, I_2)$, stands for a general 2D uncertainty (it can go in both directions so it is isotropic). I_2 is the identity matrix and multiplying it by 1.5 we get $\varepsilon \sim \mathcal{N}(0, \text{diag}(2.25, 2.25))$. The $\hat{n}_{\perp}$ term is the unit vector orthogonal to v_{plane} , so $\eta \hat{n}_{\perp}$ with $\eta \sim \mathcal{N}(0, 1)$ simulates a lateral error term. This perpendicular term $\hat{n}_{\perp} \eta$ encodes Witness 2's qualifier *"slightly off to one side"*: a symmetric perpendicular spread.

Parameters. $\alpha = 1$ corresponds to the canonical horizontal displacement implied by v_{plane} . We set $\alpha \sim \mathcal{N}(1.5, 0.5^2)$ because Witness 2 reports that the object *"fell farther along the trajectory"* (mean > 1), and Witness 1's report that it *"kept moving forward"* rules out values close to zero.

For the drift, τ has no direct witness information, so we use $\tau \sim \mathcal{N}(1.0, 0.4^2)$ centred on the canonical drift magnitude.

Under this model, Z is the sum of independent Gaussian vectors and is therefore itself Gaussian. The cell prior π_j is obtained by integrating this density over each cell. We approximate the integral by Monte Carlo sampling rather than analytic evaluation of the bivariate Gaussian CDF, since it has no closed form on rectangular cells and would become a tedious task. We sample $N = 2 \cdot 10^5$ draws and discard those outside the grid, if any.

2 Detection model

We model the conditional detection probability given coverage with a hazard function:

$$g(d_j, r_j) = \kappa \exp(-\gamma_d d_j - \gamma_r r_j),$$

where d_j and r_j are the depth and roughness respectively, at cell j . The associated penalties, $\gamma_d = 1.8$ and $\gamma_r = 1.5$ are set to these values after some calibration testing, and the fact $\gamma_r < \gamma_d$ is because the previous mission reports suggest that strong depths effects and rugged terrain severely reduce detection in this order of importance. It is a matter of choice, not any strong knowledge. Then the parameter k is the one that models how is effort going to affect the final detection probability.

We calculate the detection probability as:

$$\rho_{tj} = 1 - \exp(-e_t \cdot g(d_j, r_j)),$$

so that $q_{tj} = c_{tj} \rho_{tj}$ with $c_{tj} \in \{0, 1\}$. And so, in a perfect terrain ($d_j = d_r = 0$) with $k = 0.9$ we have probabilities 59%, 83%, 93% with respect to the three effort levels. We have consciously chosen an **imperfect detection artifact**: even in the perfect terrain with minimum effort we get 59% chance of finding the object. This is purely a design choice we have made.

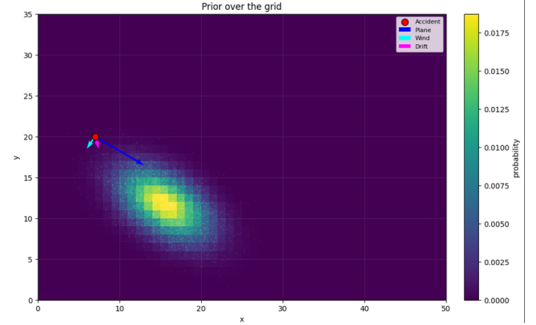


Figure 1: Prior spatial distribution.

3 Posterior update

Given a mission t over rectangle R_t with effort e_t and outcome $s_t \in \{0, 1\}$, the likelihood under the model is

$$\mathbb{P}(s_t | Z = j) = (c_{tj} \rho_{tj})^{s_t} (1 - c_{tj} \rho_{tj})^{1-s_t},$$

which, by Bayes' rule with normalising constant Z_t , gives the multiplicative update

$$\pi_j^{(t)} = \frac{1}{Z_t} \pi_j^{(t-1)} \cdot [s_t c_{tj} \rho_{tj} + (1 - s_t) (1 - c_{tj} \rho_{tj})].$$

After a non-detection ($s_t = 0$), cells outside R_t are unaffected ($c_{tj} = 0$ implies likelihood 1) and cells inside lose mass proportionally to their detectability. Crucially, cells inside R_t with high depth or roughness lose *less* mass than smooth shallow cells, because their ρ_{tj} is smaller: non-detection in hostile terrain is weak evidence, which matches intuition.

4 Mission strategy

With a budget of 530, a 12-hour cooldown and a tight deadline (I am writing this report 6 days before deadline), an effective plan must do at most 6-8 missions. So each mission must yield meaningful information. In the streamlit app, I have checked that the dollar cost is exactly the number of cells times the effort: $|R| \cdot e$. This is why we set a **minimum number** of cells per mission: $|R| \geq m$. This is key for the rectangle selection, since it reduces the space of possible missions. There is no point in small missions if the cost is proportional to the covered area and there is a large cooldown time.

We score candidate missions by their efficiency

$$\text{Eff}(R, e) = \frac{\mathbb{E}[s_t = 1]}{\text{cost}} = \frac{\sum_{j \in R} \pi_j \rho_{tj}(e)}{|R| \cdot e},$$

and implement a brute-force search over the space of all possible rectangles with a set of constraints:

1. $|R| \geq m$
2. cost > budget remaining
3. minimum_prob_detection $\geq p$

The last part of the strategy lies in how do we choose parameters. For the two first missions we set a high m ($m = 50$) and a low p ($p = 0.2$) to enforce **exploration** rather than exploitation. After the two first missions, if not found, the idea is to have a more concentrated posterior probability mass and enforce **exploitation** with lower m and higher p .

5 Results

The first mission is rectangle $[10, 20] \times [9, 15]$ with effort 2 and the bomb was **not found**. The second mission is rectangle $[12, 21] \times [8, 14]$ with effort 3 and the bomb **was found**.

We can see in the image how the posteriors after both missions.

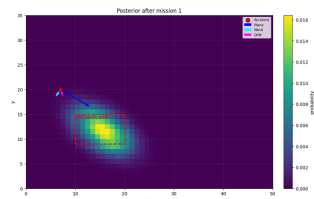


Figure 2: After Mission 1

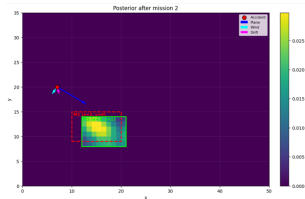


Figure 3: After Mission 2